Manual Testing using Gen AI

1. Test Planning (Before Execution)

- **Understand Requirements**
  - Analyze Business Requirement Documents (BRD) and Functional Requirement Specifications (FRS).
  - Clarify doubts with Business Analysts (BA) or Developers.
- **Create a Test Plan**
  - Define **scope, objectives, testing approach, and test deliverables**.
  - Identify required resources, timelines, and test environment needs.
  - Determine **types of testing** (Functional, Regression, Smoke, UAT, etc.).
- **Prepare Test Scenarios & Test Cases**
  - Identify **positive & negative** scenarios.
  - Write detailed **test cases** (step-by-step execution instructions).
  - Include test case attributes: **ID, Description, Preconditions, Steps, Expected Result, Priority, Status**.
  - Use a test management tool (like **JIRA, TestRail, QTest**) if available.
- **Prepare Test Data**
  - Identify required test data (e.g., usernames, passwords, input values).
  - Create sample test data using databases, Excel sheets, or predefined datasets.

2. Test Execution (During Testing)

- **Smoke Testing**
  - Perform **basic checks** to verify if the application is stable for further testing.
- **Functional Testing**
  - Execute test cases to validate each feature works as per requirements.
  - Log **bugs/defects** in a tracking tool (like **JIRA, Bugzilla**).
  - Retest fixed defects.

- **Regression Testing**
    - Ensure **new changes haven't broken existing functionality**.
    - Prioritize **critical test cases** for execution.
- **Exploratory Testing**
    - Perform **ad-hoc** testing to uncover unexpected issues.
- **UI/UX Testing**
    - Verify **design, alignment, responsiveness**, and usability.
- **Database Testing**
    - Validate data storage, updates, and retrieval using **SQL queries**.
- **Integration & API Testing** (if applicable)
    - Test **data flow** between different modules or external systems.
    - Verify APIs using **Postman/RestAssured**.

3. Test Closure (After Execution)

- **Bug Verification & Closure**
    - Verify if reported defects are fixed and working as expected.
- **Test Report Preparation**
    - Document **test execution summary** (pass/fail ratio, defects, pending tests).
- **Conduct Retrospective Meetings**
    - Discuss challenges, improvements, and learnings.
- **Sign-off & Handover**
    - Provide test artifacts to stakeholders for UAT or production release.